FUNDAMENTALS OF APPLICATION DEVELOPMENT (FAD)

Bazele dezvoltării aplicațiilor (BDA)

PBL PROJECT DEVELOPMENT | INTERNSHIP GUIDELINES

Project Description

This semester's project emphasizes foundational application development skills. Students will gain practical experience across both frontend and backend development, with a strong focus on understanding application architecture, writing robust and maintainable code, and ensuring that the final product aligns with user expectations.

In this project, students will identify and articulate essential concepts, processes, and methodologies in information processing that are critical for developing applications aligned with human activities. They will evaluate and choose suitable technologies to build software solutions that fulfil organizational requirements and effectively leverage contemporary technologies in the design and development of software applications.

This semester-long project is crafted to enable students to master the core principles of application development and to translate abstract concepts into functional solutions. The project follows a clear, iterative workflow: starting with the identification and analysis of a real-world problem, students then conduct market research to gather insights that help define both the technical and functional requirements of their solution.

With a solid foundation in place, students progress to the design and implementation phase, where they begin coding and developing prototypes that embody the essential features of their application. Throughout this process, experienced mentors provide continuous guidance, constructive feedback, and technical support to enhance the quality and effectiveness of the development.

By the end of the first 7 weeks, each team is expected to deliver a fully operational application, whether web-based, Android, or iOS, that addresses user needs and exemplifies strong engineering principles. A key focus of the project is the design and understanding of interactions among users, software interfaces, and the underlying hardware or connected devices. This hands-on experience not only equips students with technical expertise but also fosters a user-centered approach vital for crafting impactful digital solutions.

Project Deliverables

- Application architecture: grasp and implement fundamental concepts of application architecture.
- Coding practices: produce clean, efficient, and maintainable code.
- Integration: develop skills to integrate various components and services seamlessly within an application.

Project Objectives

1. Requirement analysis: gather and analyse requirements for the application.
2. Design: develop a detailed design plan, including application architecture and data models.
3. Implementation: write and integrate code for both frontend and backend components.
4. Documentation: maintain comprehensive documentation throughout the project lifecycle.
5. Presentation: prepare and deliver a final presentation demonstrating the application and discussing development experiences.

Work During the Internship

During the internship, students should work together with their company mentors on the following activities:

1. Workplace orientation: get documented about health and safety measures, emergency procedures, ergonomic recommendations, working hours, attendance, confidentiality, IT security protocols, professional behavior, communication, and teamwork.
2. Problem and domain identification: find information on potential problem domains and real-life issues and select an application topic.
3. Topic selection and feasibility: define relevance and social/technical impact; discuss scope and feasibility.
4. Project proposal: research existing solutions and background; define the problem background, relevance, and initial vision.
5. Problem analysis and domain impact: define stakeholders and needs; write the formal problem description; perform domain analysis.
6. Solution proposal and value proposition: propose the solution concept; write a user story and initial value proposition.

7. User validation preparation: define target groups and user personas; prepare interview hypotheses and draft questions.

Application Development Requirements

1. Problem and Domain Analysis

- The problem is clearly formulated.
- Target users and their needs are identified.
- The relevance and context of the problem are explained.
- The analysis supports the proposed solution.

2. Existing Solutions and Market Research

- Similar solutions are identified.
- A comparison is provided.
- Strengths and limitations are highlighted.
- The differentiation of the proposed solution is justified.

3. Functional and Non-Functional Requirements

- Requirements are clearly written.
- Functional requirements describe what the system does.
- Non-functional requirements cover performance, security, usability, etc.
- Requirements are aligned with the problem and solution.

4. Application Architecture

- The overall structure is presented.
- Main components are described.
- The relationship between frontend, backend, and database is explained.
- The architecture is coherent and justified.

5. Data Model

- Entities are correctly defined.
- Relationships are logical.
- The model supports the application functionality.
- The database structure is coherent.

6. Design Artifacts

- Required UML diagrams are included and logically correct.
- Diagrams reflect the application functionality and implementation.
- Main screens and navigation flow are presented through wireframes/mockups.
- Wireframes reflect real functionality.

7. Development and Integration

- The application runs without major errors.
- Main features are available.
- Authentication works.
- CRUD operations work correctly.
- Frontend, backend, and database components are integrated.

8. Version Control and Testing

- The GitHub repository is accessible and organized.
- Code is structured into modules/components.
- Commit history is relevant.
- Test scenarios are defined and main functionalities are tested.
- Errors are identified and fixed; results are documented.

Expected Outcomes

1. Functional application: a functional application that demonstrates the application of core development principles.
2. Component integration: smooth integration of frontend, backend, and database components.
3. Comprehensive documentation: detailed documentation covering application architecture, design decisions, code structure, and testing procedures.
4. Presentation: a final presentation showcasing the developed application, discussing the challenges faced and solutions implemented.